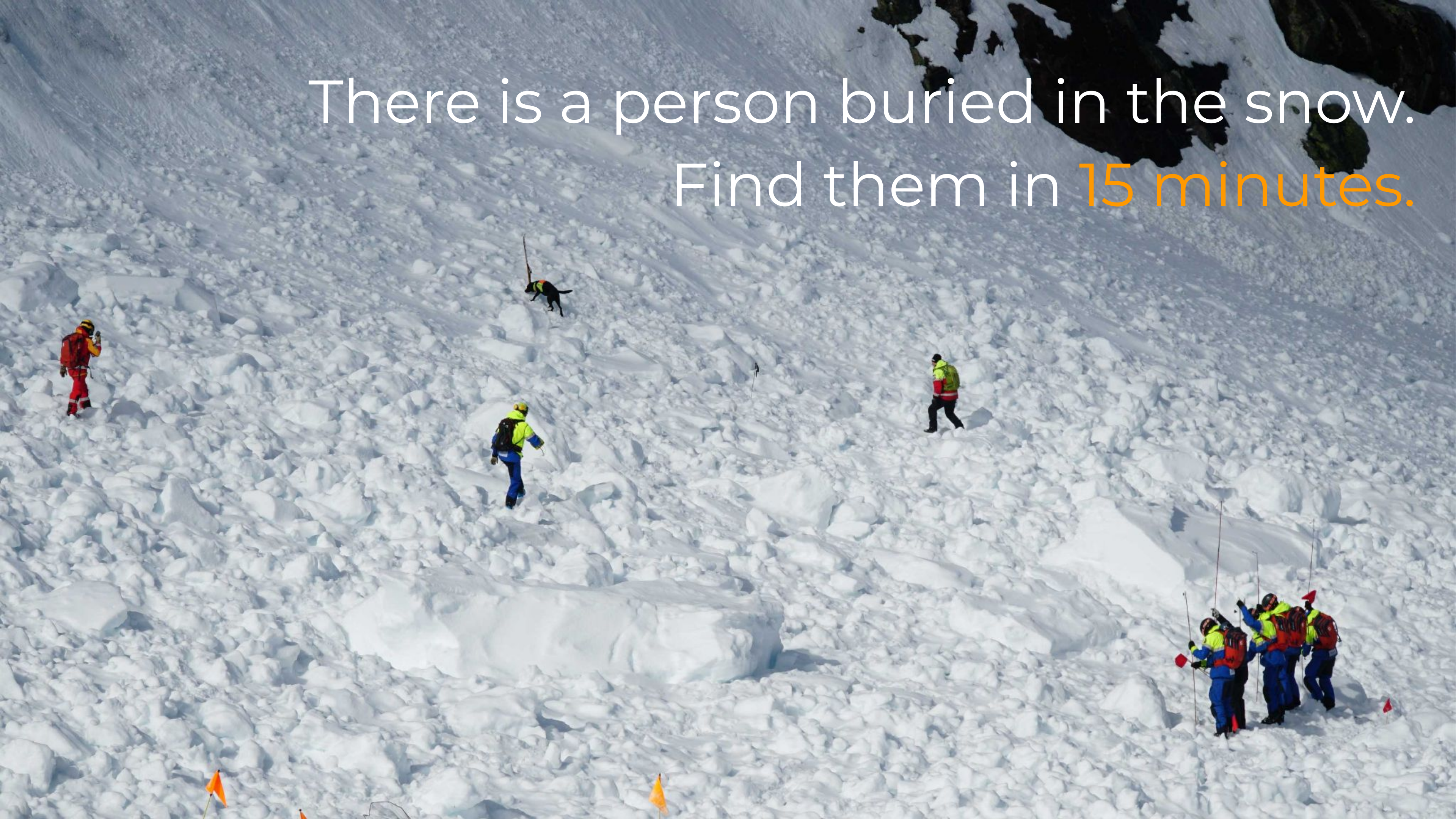
SnowBat


There is a person buried in the snow.
Find them in 15 minutes.

2,000,000+
Backcountry Skiers
in the US alone.

Companion avalanche Search-and-Rescue (SAR) Technology has largely remained the same for
60 years.

The Problem

Avalanches strike
without warning,
burying victims in
hundreds of pounds of
snow.

Survival rate:
90% → 30%
in 15 minutes.

Organized SAR is
infeasible in
backcountry scenarios:
teams are 2–6 hours
away.

Current SAR Protocol

15 min time-limit

Foot-based search using 457 kHz “Avalanche Beacon”

7 min

Manual recovery using snow probe and shovel

8 min

Avalanche Beacon

Already carried by 90% of backcountry skiers

Snow Probe

Snow Shovel

Find your friend. On foot.
You have 15 minutes.

Image modified with Google Gemeni

SnowBat

Purpose-built drone for avalanche SAR

Autonomously finds the buried victim in minutes

One button press.
One life saved.

Deploy SnowBat:
1. Orient SnowBat toward last known position
2. Press launch

Autonomous Search:
1. Sweeps debris field for beacon signal
2. Locks onto victim's 457 kHz transmission
3. Guides rescuers directly to burial site

Manual Extraction:
1. Extricate victim with snow shovel

Images are AI generated using Google Gemini using a CAD model of SnowBat

Mechanical
Design

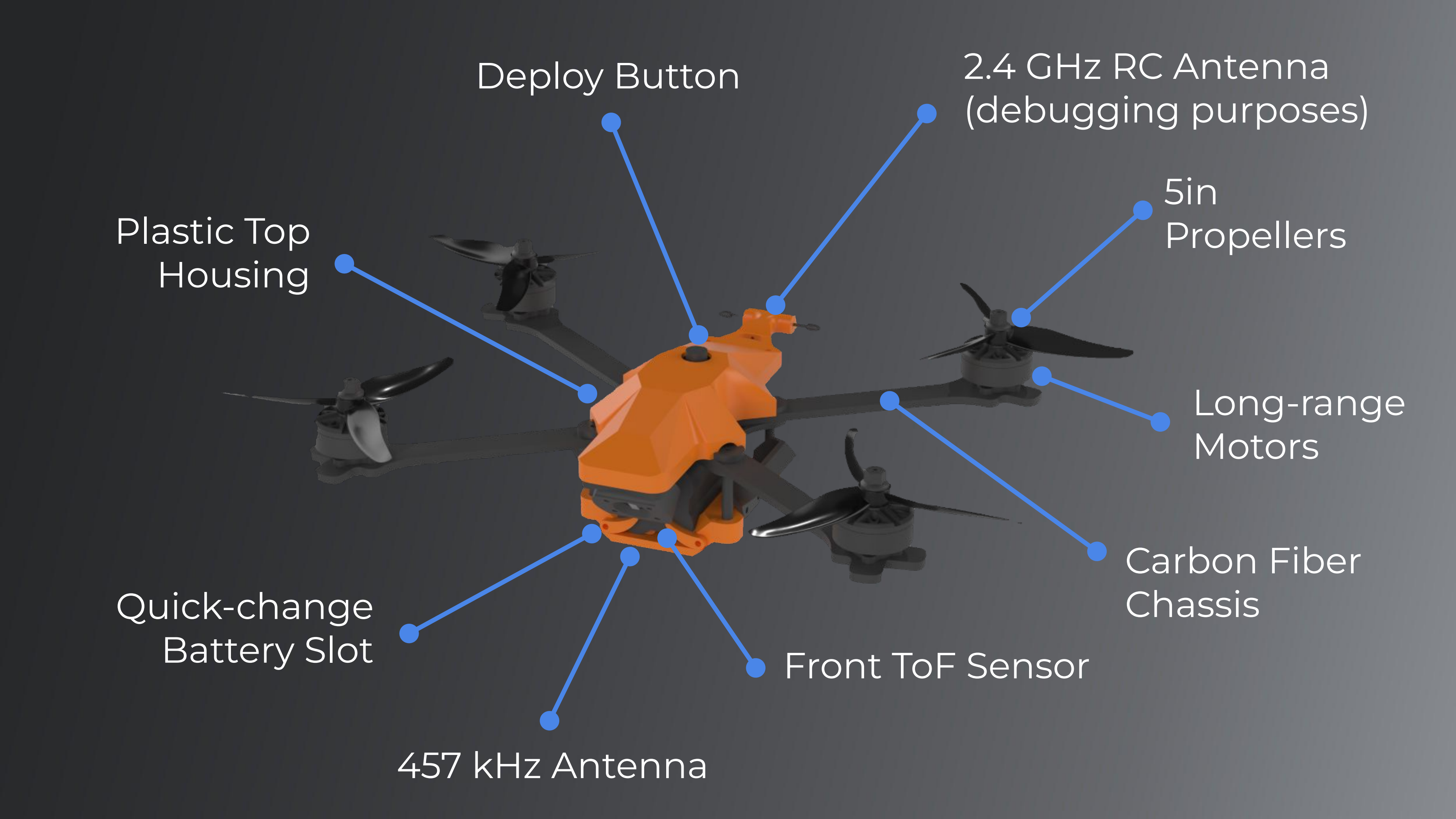
Design Goals & Considerations

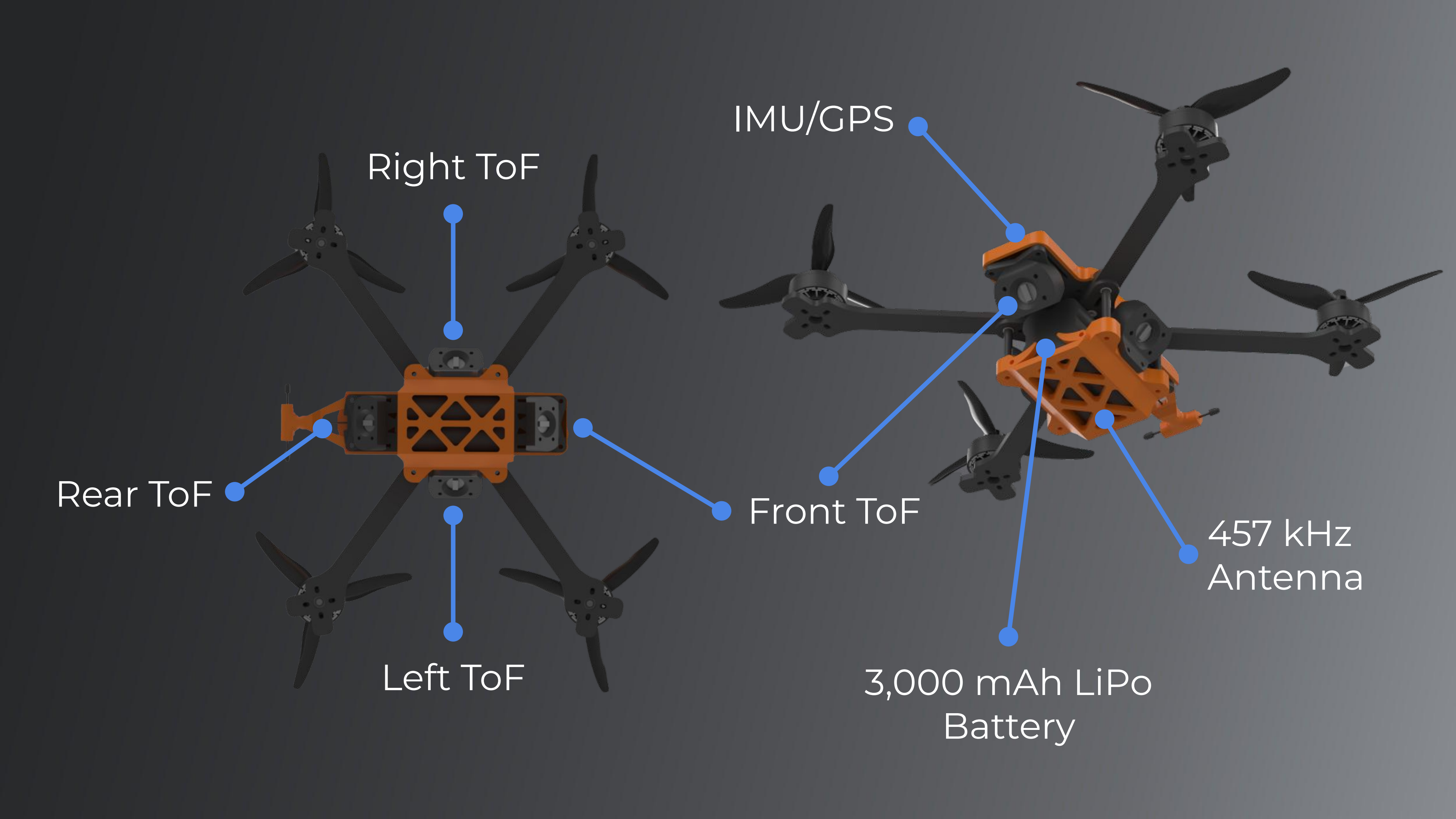
Mass:	1.5 lbs (0.68 kg)
Endurance:	20 minutes
Wind Tolerance:	Up to 35 MPH, gale-force winds
Altitude:	10,000 ft (3048 m)
Form Factor:	Collapsible and packable

Lightweight.
Packable.
Backpack-ready.

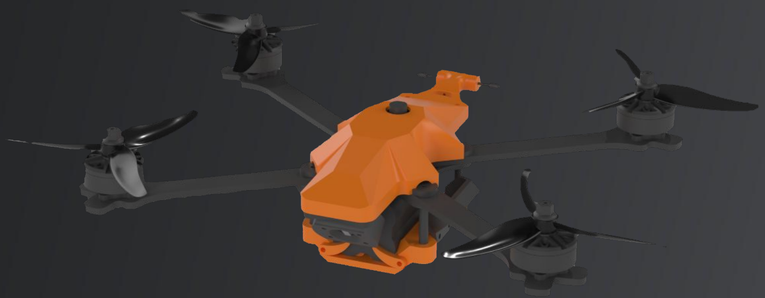
1.5 lbs
Collapsible Design
Deployed Size: 11.5in x 14.5in
Collapsed Size: 7.0in x 10in

Weight and size are comparable to a 1L water bottle.


Deploy Button
2.4 GHz RC Antenna (debugging purposes)
5in Propellers
Plastic Top Housing
Long-range Motors
Carbon Fiber Chassis
Quick-change Battery Slot
Front ToF Sensor
457 kHz Antenna

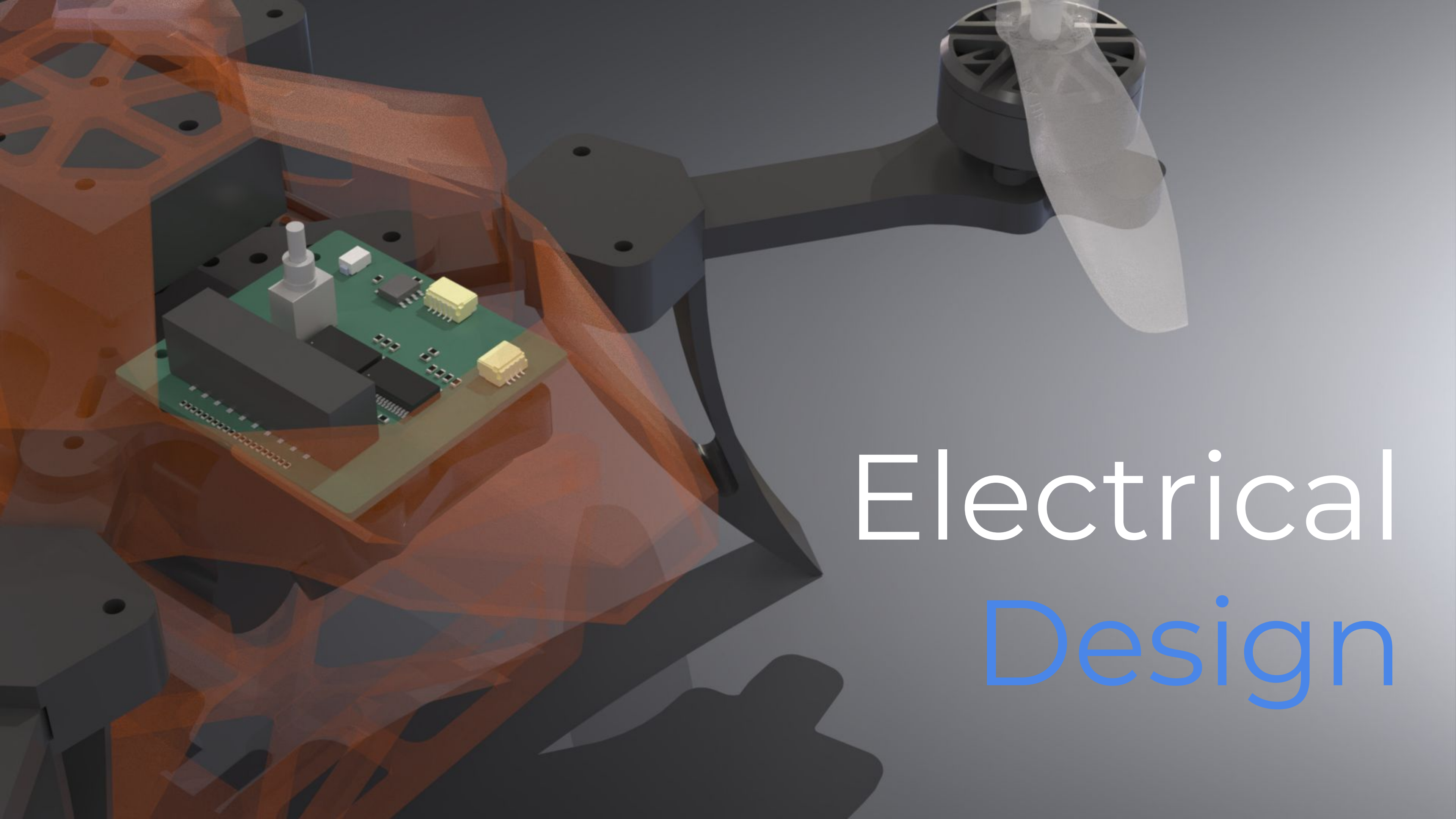

Right ToF
Rear ToF
Left ToF
IMU/GPS
Front ToF
457 kHz
Antenna
3,000 mAh LiPo
Battery

Evolution of Mechanical Design



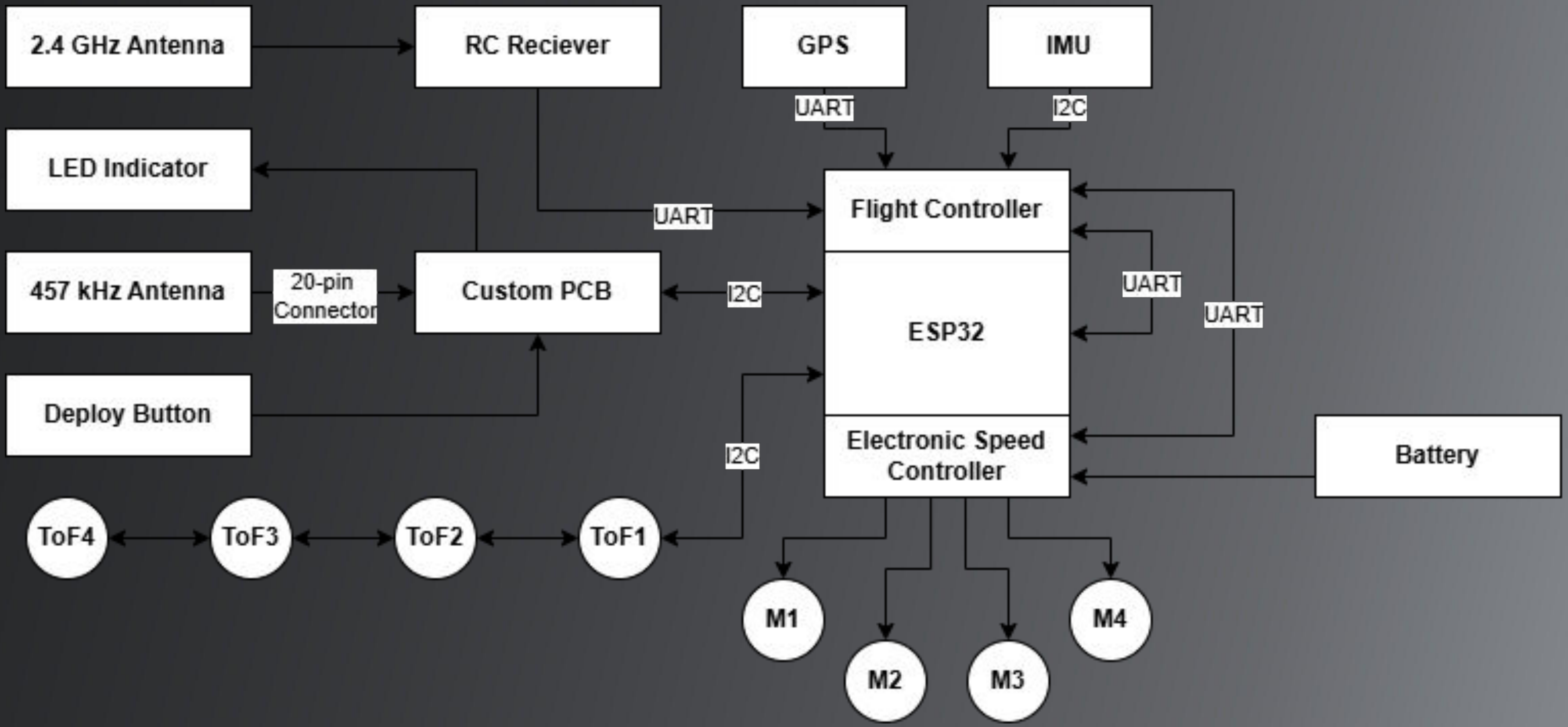
SnowBat v1

SnowBat v1
Test Bench
Prototype



Electrical Design

Systems Diagram



Electronics & Sensor Selection

- Battery
    - 2S, 7.4V
    - 3000mAh
- ToF Sensors
    - ~3 feet range, 66mW (20mA at 3v3), I2C
    - Verified to work on snow
- BCA Avalanche Beacon
    - Highly standardized
    - Transmit/receive 457 kHz signals at once per second
    - 50 meter range + direction

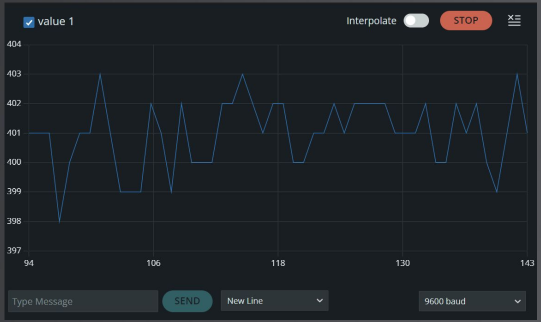
ToF Sensor Testing

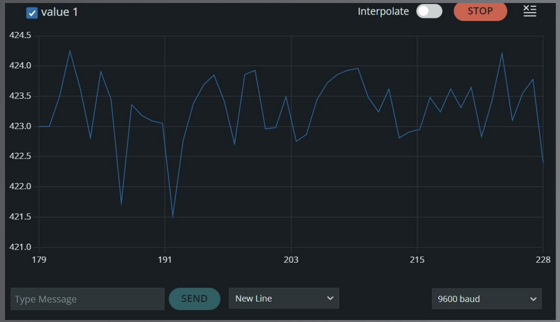
Strong drop-off in performance at ~800 mm

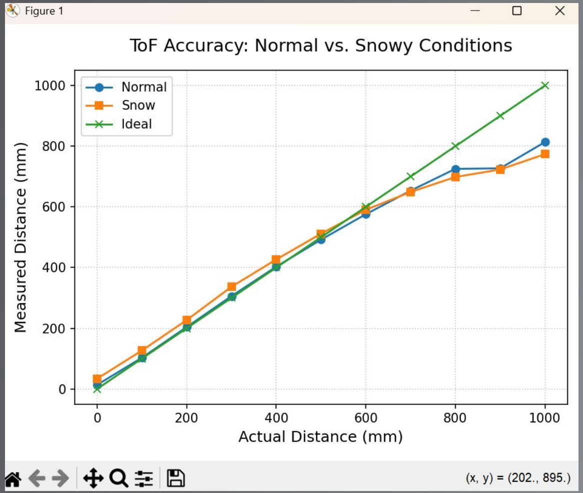
- 15 mm offset
- In the snow, ~30 mm offset
- Snow R^2: 0.9091
- Normal R^2: 0.9335

For <800 mm:

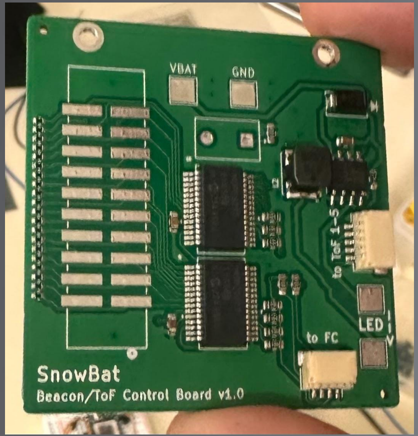
- Snow: 0.9823
- Normal: 0.9926





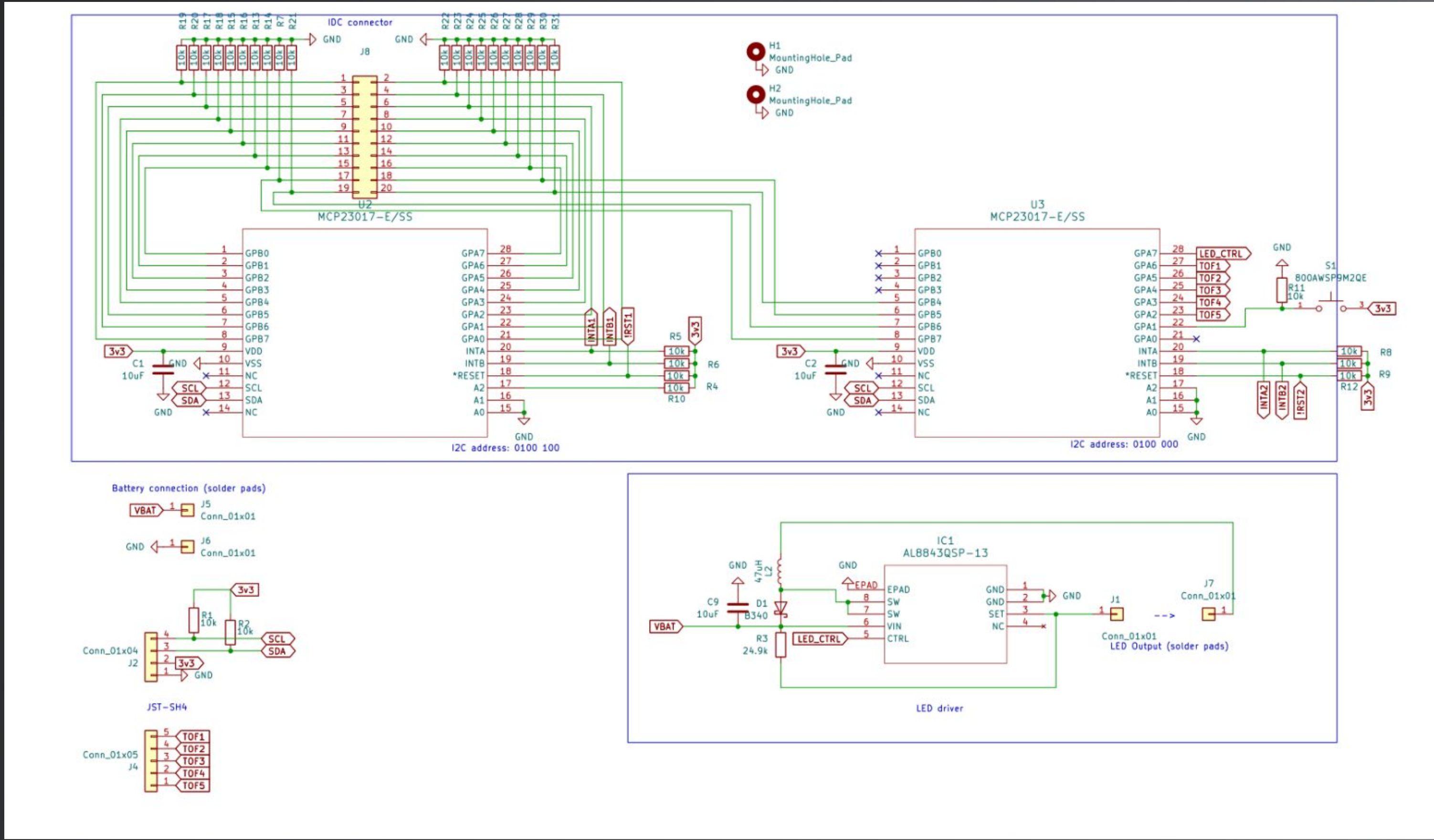


PCB Design Goals & Considerations



Function:	Read 20 pins from the beacon display Control 4 ToF sensors Activation button and LED
Form Factor:	Fits inside the existing drone form factor

Custom PCB Schematic



PCB Key Features and Components

MCP23017 – GPIO Expander

- I2C 16-pin I/O expansion
- Internal pull-up resistors

AL8843QSP-13 – LED Driver

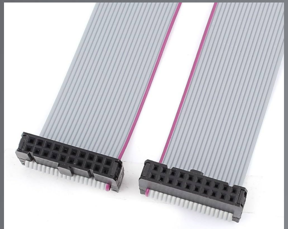
- Drives a 3W LED at 600 mA
- Estimated ~1 million nits without diffuser

Button

- IP67-rated for environmental protection
- Ergonomic, tall-profile design for external drone mounting
- Built for reliable, long-lasting operation

IDC Connector

- Compact and low-profile
- Flexible routing for tight assemblies
- Speeds up cable assembly

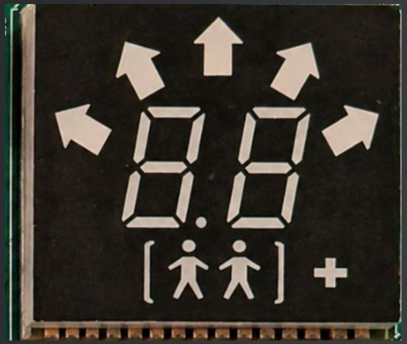


Decoding the Beacon

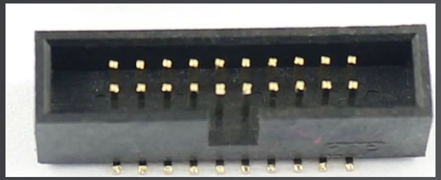
LED display pins→ IDC connector pins→ MCP GPIO pins

One-hot encoded

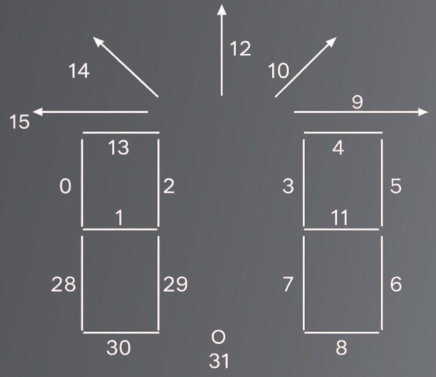
Based on a clockwise numbering of the 7-segment displays



Beacon



Custom PCB



Final mapping in software

Software
Architecture

Firmware

High-level testing and basic flight validation

- BetaFlight

More custom integration with our various devices (time of flights, beacon) over I2C

- Ardupilot: https://github.com/Ardupilot/ardupilot
- Specifically, the ArduCopter codebase

Time of Flight (ToF) Sensors

XSHUT is an enable/reset pin. Pull all XSHUT pins low, turn on one sensor & change its I2C address, repeat

Each sensor has

- 1.2 meter range
- 45° angle from the x-y plane

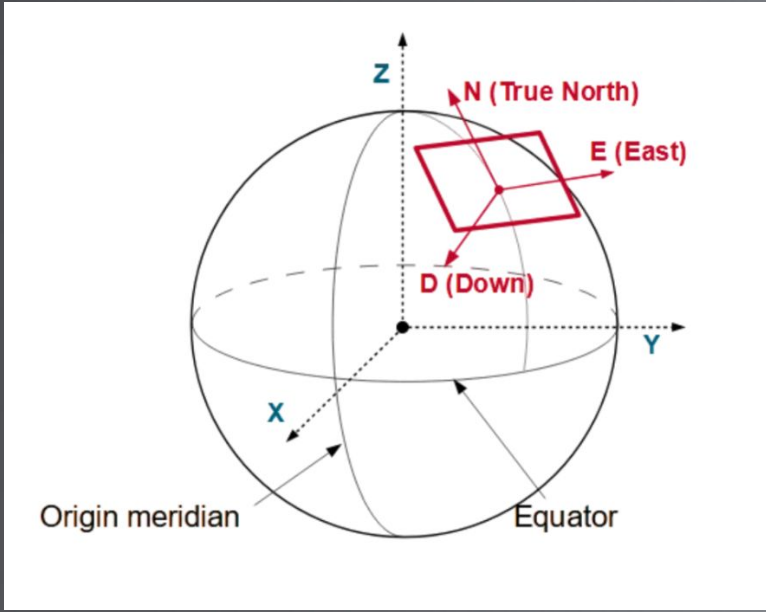
$$h_i = d_i \cdot \sin\theta$$

$$h = \sum d_i \cdot \sin\theta = \sin\theta \cdot \frac{d_1 + d_2 + d_3 + d_4}{4}$$

Max hover height of 1.2m * sin(45°) = 2.78 ft

ArduPilot Waypoints and Coordinate System

- The built-in flight mode, Guided, provides autonomous waypoint navigation
- Accepts North-East-Down (NED) coordinates



a localized "tangent plane" to the WGS84 ellipsoid

Waypoint Navigation

Beacon gives us a direction and approximate distance to target

Given:

- current NED coordinate
- current absolute yaw
  - estimated by an EKF using the Magnetometer, Gyroscope, and GPS

We convert this one-hot-encoded direction to a desired yaw angle

```
Vector3f beacon_reading_to_ned_coord(const Vector3f& current_ned,
                                     float drone_yaw_rad,
                                     float direction_rad,
                                     float distance_m,
                                     float max_step_m)
{
    // Cap the step size. This is especially useful during the beginning
    const float step_m = min(x: distance_m, y: max_step_m);

    // Convert body-relative direction to world/NED direction
    const float world_direction_rad = drone_yaw_rad + direction_rad;

    Vector3f target;
    target.x = current_ned.x + step_m * cosf(X: world_direction_rad); // North
    target.y = current_ned.y + step_m * sinf(X: world_direction_rad); // East
    target.z = current_ned.z;  // Down unchanged here... modified later with ToF data

    return target;
}
```

Waypoint Generation Logic

Currently...

Loop:

- Command drone to move to certain NED coordinate along given beacon direction (non-blocking)
- While L2 norm between current and target NED coordinate is above a defined tolerance, do nothing

Working towards...

Gradient descent towards victim... go in direction of steepest decrease in distance

Why?

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8}{4.57 \cdot 10^5} = 650m$$

- Beacon antenna is only a few centimeters long → electrically tiny (<< λ)
- Behaves like magnetic dipole field. Flux lines form closed loops like a bar magnet

Single-button deployment.

Integrates into existing SAR protocol. Works with every avalanche beacon.

Fully Autonomous.

SnowBat

1.5 lbs. Collapsible. Backpack-ready.

The first-ever purpose-built, companion-deployable drone for avalanche SAR.

Rugged. Reliable.

Save Lives.

Choose Safety.
Choose SnowBat.